

Knowledge Compilation

Outline

1 How to model using SAT

2 Knowledge Compilation

- Knowledge Compilation Map (Darwiche and Marquis, 2002)
 - Queries and Transformations
 - Succinctness
- SAT Based Compilers
- Top-Down Compilers
- Dynamic Programming and Compilation
- Bottom-Up Compilers

3 d-DNNF Queries

4 Conclusion and References

Motivations for Knowledge Compilation

- **The Bottleneck:** SAT and #SAT are NP-complete and #P-complete. In practice, there is no guarantee of solving queries within a strict time limit.

The KC Paradigm: Split the Effort

- 1 **Offline Phase (Compilation):** Translate the instance Σ into a representation $\mathcal{C}$ from a target language $\mathcal{L}$. This step is computationally expensive.
 - 2 **Online Phase (Querying):** Use $\mathcal{C}$ to answer difficult queries (like Model Counting) in polynomial time.
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- **When is it useful?** When the offline cost is amortized over *many* queries on the same fixed knowledge base.
 - **The Question:** Which language $\mathcal{L}$ should we choose? We use the Knowledge Compilation Map.

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KC Map: Evaluating a Language $\mathcal{L}$

To choose a language, we evaluate its tractability on two axes:

1. Queries

(Extracting Information)

- **CO** (Consistency / SAT)
- **CE** (Clause Entailment)
- **VA** (Validity)
- **EQ** (Equivalence)
- **SE** (Sentential Entailment)
- **IM** (Implicants)
- **CT** (Model Counting)
- **ME** (Model Enumeration)

2. Transformations

(Modifying the Knowledge)

- **CD** (Conditioning)
- $\wedge \mathbf{C}, \wedge \mathbf{BC}$ (Closure under $\wedge$)
- $\vee \mathbf{C}, \vee \mathbf{BC}$ (Closure under $\vee$)
- $\neg \mathbf{C}$ (Closure under $\neg$)
- **FO**, **SFO** (Forgetting)

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- $\neg\mathbf{C}$ (Closure under $\neg$)
- **FO, SFO** (Forgetting)

Target Languages: The Baselines

Before evaluating their properties, let us define our starting languages.

Circ (Boolean Circuits)

A Directed Acyclic Graph (DAG) where internal nodes are Boolean gates ($\wedge, \vee, \neg$) and leaves are variables or constants.

- **Pros:** Can represent any Boolean function compactly.
- **Cons:** Almost all queries (SAT, #SAT) remain NP-hard / #P-hard.

CNF

An AND of ORs (clauses).

- **Ex:** $(a \vee \neg b) \wedge (\neg a \vee c)$
- Great for modeling (like our Zykov encoding).
- **CO** (Consistency) is hard.

DNF

An OR of ANDs (terms).

- **Ex:** $(a \wedge \neg b) \vee (\neg a \wedge c)$
- Explicitly lists partial models.
- **CO** is easy ($\checkmark$), but **CT** (Counting) is still hard!

Fragment of the KC Map

- $\checkmark$: Polynomial-time algorithm exists.
- $\circ$: No poly-time algorithm (unless $P = NP$).

Queries

$\mathcal{L}$	CO	VA	CE	IM	EQ	SE	CT	ME
Circ	$\circ$	$\circ$	$\circ$	$\circ$	$\circ$	$\circ$	$\circ$	$\circ$
CNF	$\circ$	$\checkmark$	$\circ$	$\checkmark$	$\circ$	$\circ$	$\circ$	$\circ$
DNF	$\checkmark$	$\circ$	$\checkmark$	$\circ$	$\circ$	$\circ$	$\circ$	$\checkmark$

Transformations

$\mathcal{L}$	CD	FO	SFO	$\wedge C$	$\wedge BC$	$\vee C$	$\vee BC$	$\neg C$
Circ	$\checkmark$	$\circ$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
CNF	$\checkmark$	$\circ$	$\checkmark$	$\checkmark$	$\checkmark$	$\circ$	$\checkmark$	$\circ$
DNF	$\checkmark$	$\checkmark$	$\checkmark$	$\circ$	$\checkmark$	$\checkmark$	$\checkmark$	$\circ$

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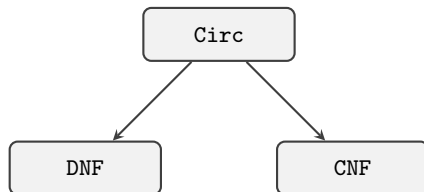
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Succinctness ($\leq_s$)

Succinctness captures the ability of a language to represent information **compactly**.

- $\mathcal{L}_1 \leq_s \mathcal{L}_2$ ($\mathcal{L}_1$ is *at least as succinct as* $\mathcal{L}_2$) if there exists a polynomial p such that for every formula $\alpha \in \mathcal{L}_2$, there is an equivalent $\beta \in \mathcal{L}_1$ where $|\beta| \leq p(|\alpha|)$.
- An arrow $\mathcal{L}_1 \rightarrow \mathcal{L}_2$ strictly means $\mathcal{L}_1 <_s \mathcal{L}_2$ (Circ is strictly more succinct than the others)



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Leveraging Modern SAT Solvers

- Over the last two decades, SAT solvers (CDCL) have become incredibly efficient at finding a single satisfying assignment for massive CNF formulas.
- **The Question:** Instead of building a custom compilation algorithm from scratch, can we treat an off-the-shelf SAT solver as an *oracle* to compile our formula?

The Black-Box Compilation Approach

Iteratively query the SAT solver. When it finds a solution, record it, *block it* from the search space, and query the solver again until the formula becomes unsatisfiable.

- **What target languages does this produce?**
 - If we extract full assignments (models), we generate MODS.
 - If we extract partial assignments (prime implicants), we generate ODNF.
- To understand this approach, let us first review how modern CDCL SAT solvers work.

The Resolution Rule

- **The Rule:** Two clauses containing a variable x with *clashing literals* ($x, \neg x$) imply a *resolvent* clause containing the union of the remaining literals.

$$\frac{\underbrace{x \vee y \vee \neg z}_{C_1} \quad \underbrace{\neg x \vee t \vee u}_{C_2}}{y \vee \neg z \vee t \vee u}$$

- **Intuition:** If the formula is true, x must be either True or False. In either case, one of the remaining parts (C_1 or C_2) must hold.
- **Variable Elimination:** Performing all possible resolutions on x is equivalent to *forgetting* x (existential quantification):

$$\exists x. \Sigma \equiv (\Sigma|_x) \vee (\Sigma|_{\neg x})$$

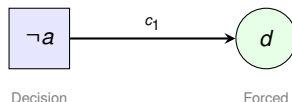
- **Property:** Resolution is a refutation-complete proof system for CNF formulas.

Ingredients of a CDCL Solver

■ 1. Decisions & Propagation

- **Branching Heuristics:** Choosing the next variable (e.g., VSIDS) and its polarity.
- **Unit Propagation (BCP):** Deduce values forced by clauses.

$$\phi = \{c_1 : \neg a \vee d\}$$



■ 2. Conflict Analysis & Learning

- Building the *Implication Graph* upon conflict.
- *Clause Learning*: Deriving a new clause to prevent this mistake.
- *Backjumping*: Non-chronological backtracking.

Key Mechanism

Constructing the **Implication Graph** allows the solver to learn from failure.

Constructing the Implication Graph

Current Formula State

$$\begin{array}{lll} c_1 : a \vee d & c_2 : a \vee \neg c \vee \neg f & c_3 : \neg d \vee j \vee f \\ c_4 : b \vee h & c_5 : \neg c \vee \neg e \vee i & c_6 : \neg i \vee \neg j \vee \neg g \\ c_7 : e \vee \neg k & c_8 : e \vee \neg h \vee k & c_9 : \neg c \vee \neg e \vee \neg i \vee g \end{array}$$

Decisions & Propagation Trace:

DL 1

DL 2

DL 3

DL 4

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$$\neg a$$

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Decisions & Propagation Trace:

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DL 2

DL 3

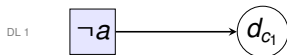
DL 4

Constructing the Implication Graph

Current Formula State

$$\begin{array}{lll} c_1 : a \vee d & c_2 : a \vee \neg c \vee \neg f & c_3 : \neg d \vee j \vee f \\ c_4 : b \vee h & c_5 : \neg c \vee \neg e \vee i & c_6 : \neg i \vee \neg j \vee \neg g \\ c_7 : e \vee \neg k & c_8 : e \vee \neg h \vee k & c_9 : \neg c \vee \neg e \vee \neg i \vee g \end{array}$$

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DL 2

DL 3

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$c_1 : a \vee d$	$c_2 : a \vee \neg c \vee \neg f$	$c_3 : \neg d \vee j \vee f$
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Decisions & Propagation Trace:



DL 3

DL 4

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Decisions & Propagation Trace:



DL 3

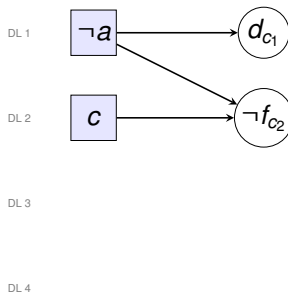
DL 4

Constructing the Implication Graph

Current Formula State

$c_1 : a \vee d$	$c_2 : a \vee \neg c \vee \neg f$	$c_3 : \neg d \vee j \vee f$
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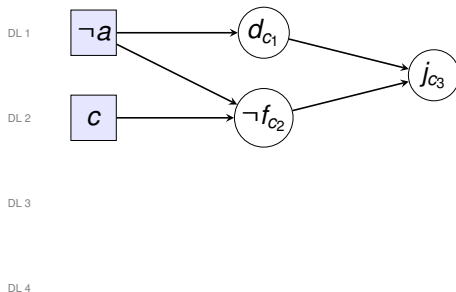


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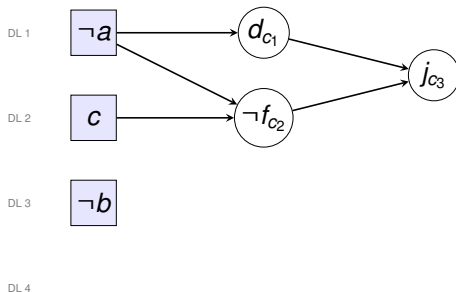


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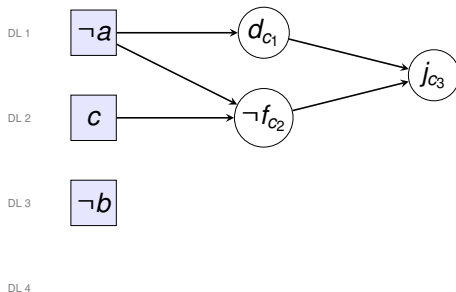


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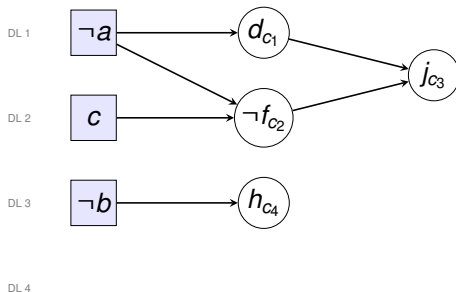


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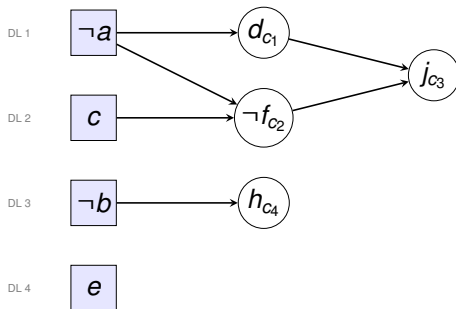


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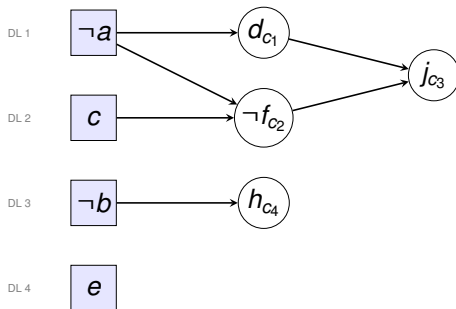


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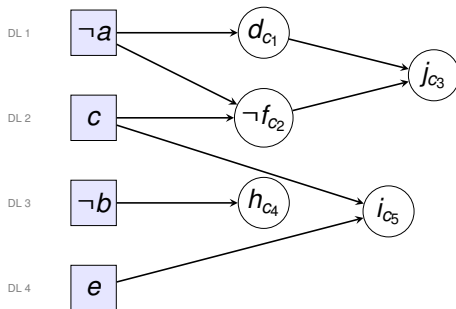


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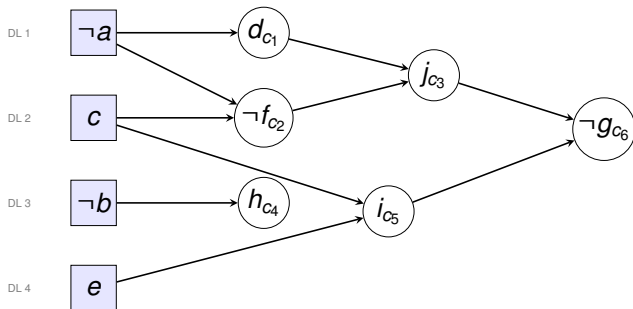


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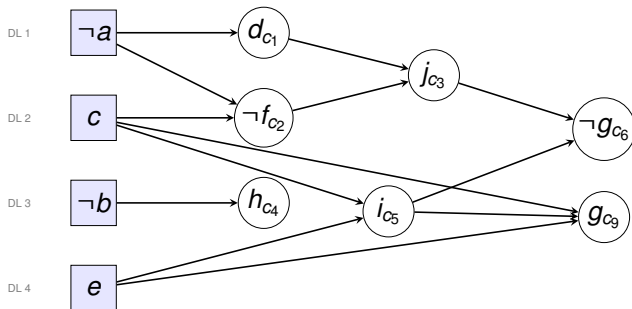


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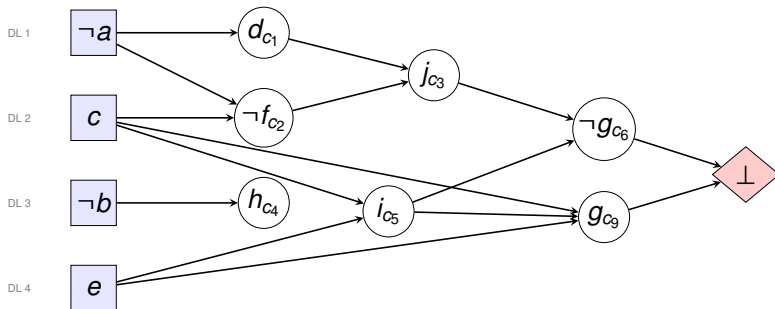


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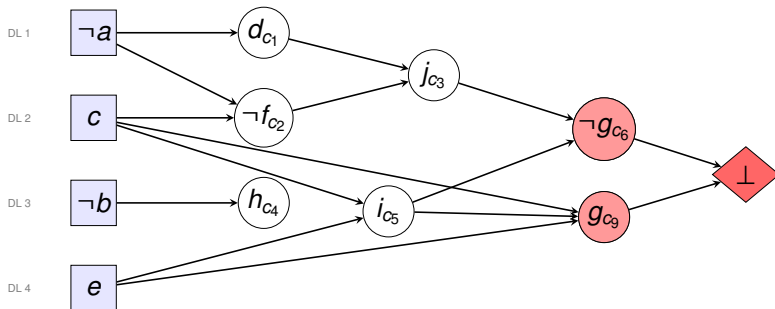


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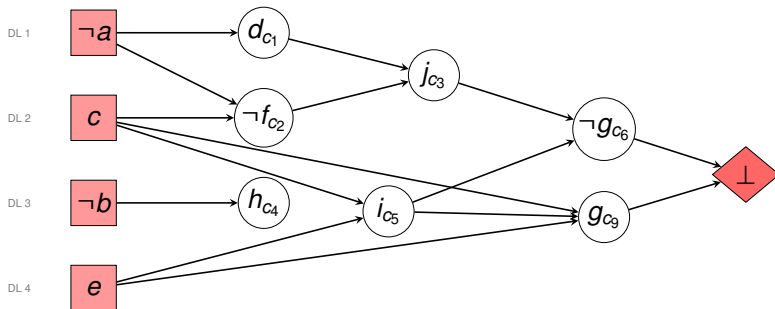


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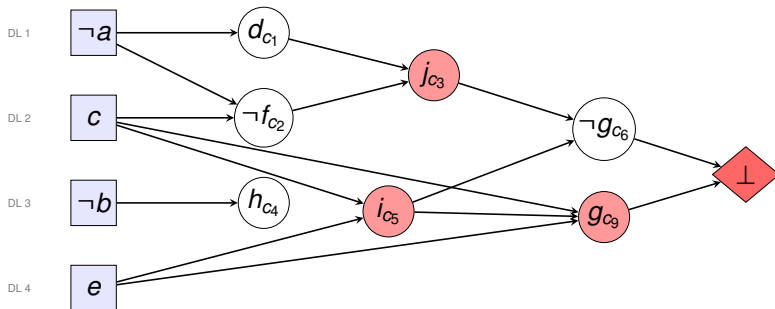


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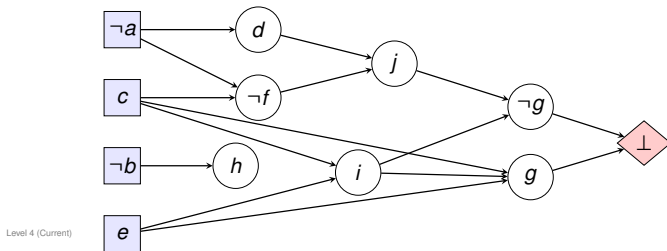
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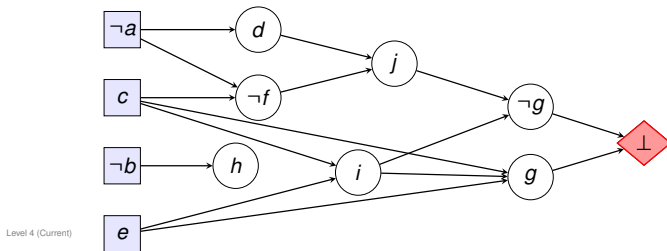
Conflict Analysis & Clause Learning



Resolution Trace (Deriving δ)

Conflict detected at level 4.

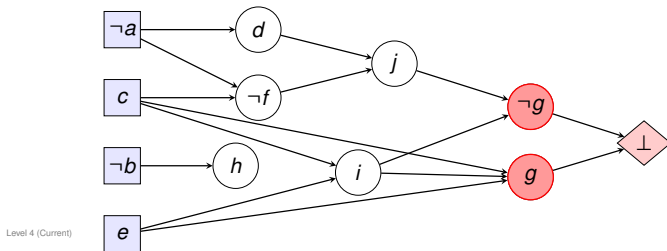
Conflict Analysis & Clause Learning



Resolution Trace (Deriving δ)

Start with conflict: $\perp$

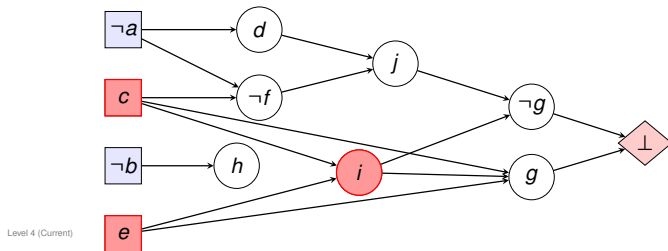
Conflict Analysis & Clause Learning



Resolution Trace (Deriving δ)

Resolve on g : $g \vee \neg g$

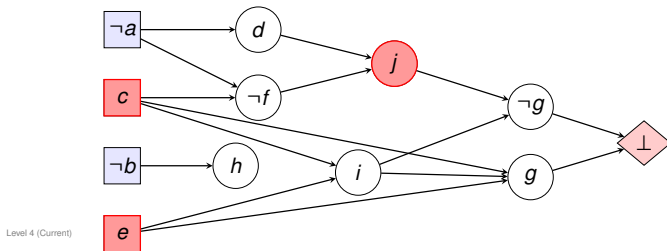
Conflict Analysis & Clause Learning



Resolution Trace (Deriving δ)

Resolve inputs of g : $\neg c \vee \underbrace{\neg e \vee \neg i}_{\text{Level 4}} \vee \neg j = \neg i \vee \neg j \vee \neg g \oplus \neg c \vee \neg e \vee \neg i \vee g$
 (Two literals at Level 4: e, i . **Continue.**)

Conflict Analysis & Clause Learning

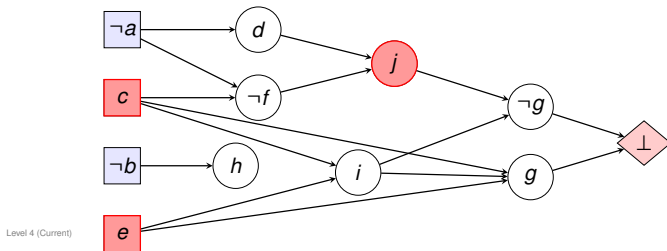


Resolution Trace (Deriving δ)

Resolve on i : $\neg c \vee \neg e \vee \neg j = \neg c \vee \neg e \vee \neg i \vee \neg j \oplus \neg c \vee \neg e \vee i$
(**Stop**: Only one literal at Level 4: e)

- **1-UIP Condition:** Stop when the resolvent contains exactly *one literal* from the current decision level (here e).

Conflict Analysis & Clause Learning



Resolution Trace (Deriving δ)

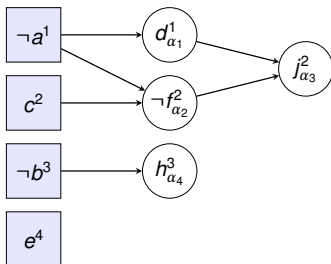
Resolve on i : $\neg c \vee \neg e \vee \neg j = \neg c \vee \neg e \vee \neg i \vee \neg j \oplus \neg c \vee \neg e \vee i$
(**Stop**: Only one literal at Level 4: e)

- **1-UIP Condition:** Stop when the resolvent contains exactly *one literal* from the current decision level (here e).
- **Learning:** Add clause $\delta = (\neg c \vee \neg e \vee \neg j)$ to the database.

Backjumping

Current Formula State

$\alpha_1 : a \vee d$ $\alpha_2 : a \vee \neg c \vee \neg f$ $\alpha_3 : \neg d \vee j \vee f$
 $\alpha_4 : b \vee h$ $\alpha_5 : \neg c \vee \neg e \vee i$ $\alpha_6 : \neg i \vee \neg j \vee \neg g$
 $\alpha_7 : e \vee \neg k$ $\alpha_8 : e \vee \neg h \vee k$ $\alpha_9 : \neg c \vee \neg e \vee \neg i \vee g$

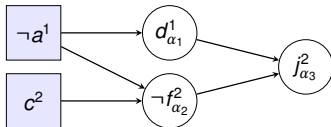


$$\delta_1 = \neg c^2 \vee \neg j_{\alpha_3}^2 \vee \neg e^4$$

Backjumping

Current Formula State

$\alpha_1: a \vee d$	$\alpha_2: a \vee \neg c \vee \neg f$	$\alpha_3: \neg d \vee j \vee f$
$\alpha_4: b \vee h$	$\alpha_5: \neg c \vee \neg e \vee i$	$\alpha_6: \neg i \vee \neg j \vee \neg g$
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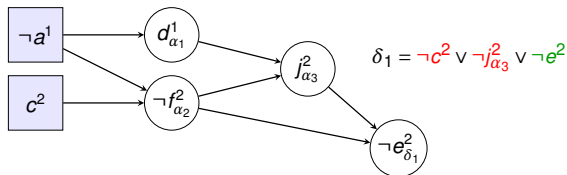


$$\delta_1 = \neg c^2 \vee \neg j_{\alpha_3}^2 \vee \neg e$$

Backjumping

Current Formula State

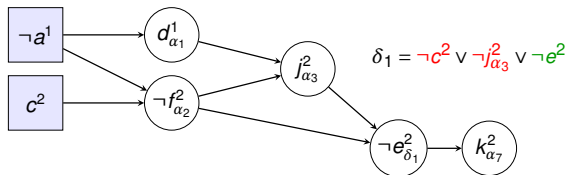
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Backjumping

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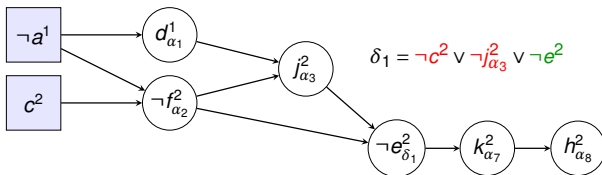
$\alpha_1 : a \vee d$ $\alpha_2 : a \vee \neg c \vee \neg f$ $\alpha_3 : \neg d \vee j \vee f$
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Backjumping

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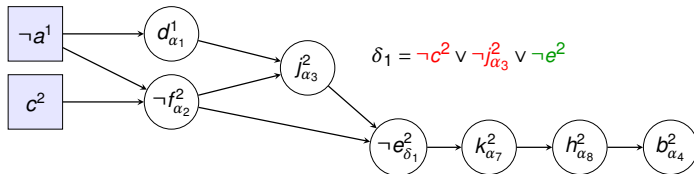
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Backjumping

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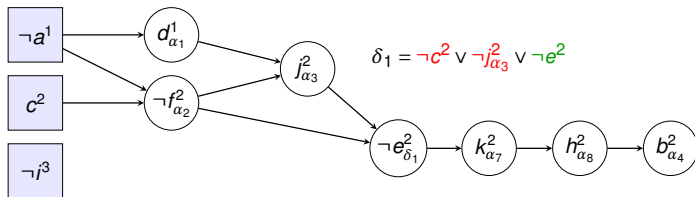
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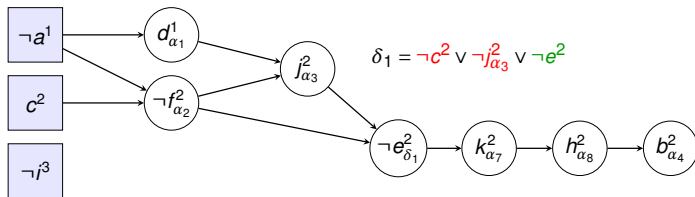
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SATISFIABILITY PROVED

Outline

1 How to model using SAT

2 Knowledge Compilation

- Knowledge Compilation Map (Darwiche and Marquis, 2002)

■ SAT Based Compilers

- MODS

- ODNF

- Top-Down Compilers

- Dynamic Programming and Compilation

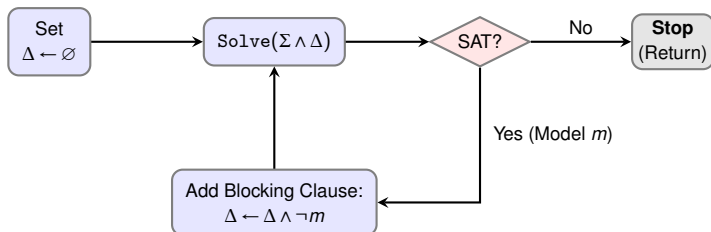
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3 d-DNNF Queries

4 Conclusion and References

Generating MODS: The All-SAT Loop

- A naive Knowledge Compilation language is MODS: explicitly listing all satisfying assignments.
- We can compute this incrementally using a standard SAT solver (Audemard et al., 2013).



- *Note:* A blocking clause $\neg m$ rules out the model m so the solver is forced to find a new one.

KC Map for MODS: Queries

$\mathcal{L}$	CO	VA	CE	IM	EQ	SE	CT	ME
Circ	○	○	○	○	○	○	○	○
CNF	○	✓	○	✓	○	○	○	○
DNF	✓	○	✓	○	○	○	○	✓
MODS	✓	✓	✓	✓	✓	✓	✓	✓

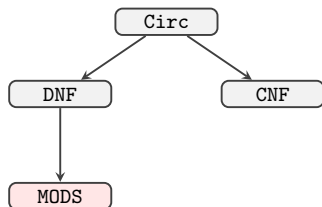
- **Notice:** MODS is the only language here that trivially supports every single query in polynomial time.

KC Map for MODS: Transformations

$\mathcal{L}$	CD	FO	SFO	$\wedge C$	$\wedge BC$	$\vee C$	$\vee BC$	$\neg C$
Circ	✓	○	✓	✓	✓	✓	✓	✓
CNF	✓	○	✓	✓	✓	○	✓	○
DNF	✓	✓	✓	○	✓	✓	✓	○
MODS	✓	✓	✓	○	✓	○	○	○

- **The Trade-off:** While its queries are trivial, MODS loses significant transformation capabilities compared to general circuits (e.g., no closure under general $\wedge$ or $\vee$).

The Achilles Heel of MODS: Succinctness



Why is MODS strictly less succinct than DNF?

A single term in DNF (e.g., x_1) can implicitly represent 2^{n-1} models. MODS forces us to list them all.

A Disastrous Example:

Can we compile this simple formula efficiently into MODS?

$$\Sigma = \bigvee_{i=1}^n x_i$$

- Size in Circ/ CNF / DNF: $\mathcal{O}(n)$
- Size in MODS: $2^n - 1$ models!

Conclusion: MODS provides ultimate tractability, but zero compression. We need an intermediate language.

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Between MODS and DNF: Orthogonal DNF (ODNF)

- DNF is succinct but fails at Model Counting (**CT**) because terms can overlap.
- MODS makes **CT** easy, but has zero compression (lists every single assignment).

Orthogonal DNF (ODNF)

A formula in DNF ($\bigvee_{i=1}^m t_i$) where all terms are **mutually exclusive** (orthogonal):

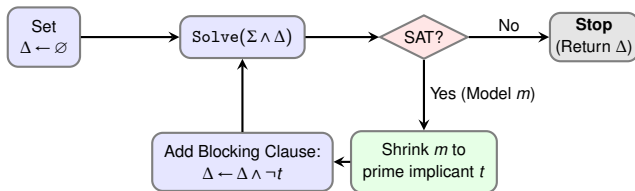
$$t_i \wedge t_j \models \perp \quad \text{for all } i \neq j$$

Why it works for Model Counting:

- Because terms do not overlap, the total number of models is exactly the sum of the models of each term.
- A single term t of length k (over n variables) compactly represents 2^{n-k} models.
- *Example:* If $n = 10$, the term $(x_1 \wedge \neg x_2)$ represents $2^8 = 256$ models perfectly, replacing 256 lines of MODS!

Compiling ODNF from CNF

We can adapt our MODS generator. Instead of blocking a full model m , we shrink m into a **Prime Implicant** t and block that instead! (Audemard et al., 2013; Spallitta et al., 2024)



- **Why is it Orthogonal?** The clause $\neg t$ forces the next implicant to contradict t .
- **Why is shrinking fast?** Because Σ is in CNF! Checking if a reduced term t is still an implicant ($t \models \Sigma$) simply requires checking if t contains at least one literal from every clause in Σ .

The KC Map: Queries for ODNF

$\mathcal{L}$	CO	VA	CE	IM	EQ	SE	CT	ME
Circ	○	○	○	○	○	○	○	○
CNF	○	✓	○	✓	○	○	○	○
DNF	✓	○	✓	○	○	○	○	✓
MODS	✓	✓	✓	✓	✓	✓	✓	✓
ODNF	✓	✓	✓	✓	✓	✓	✓	✓

- **The Breakthrough:** By simply forcing the terms of a DNF to be mutually exclusive, Model Counting (CT) flips from ○ to ✓!
- *Why?* Because $\|t_1 \vee t_2\| = \|t_1\| + \|t_2\|$ is only mathematically true if $t_1 \wedge t_2 \models \perp$.

The KC Map: Transformations for ODNF

$\mathcal{L}$	CD	FO	SFO	$\wedge C$	$\wedge BC$	$\vee C$	$\vee BC$	$\neg C$
Circ	✓	○	✓	✓	✓	✓	✓	✓
CNF	✓	○	✓	✓	✓	○	✓	○
DNF	✓	✓	✓	○	✓	✓	✓	○
MODS	✓	✓	✓	○	✓	○	○	○
ODNF	✓	○	○	○	✓	○	○	○

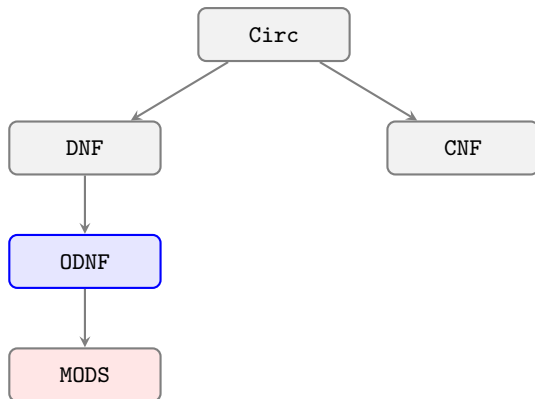
Why is Singleton Forgetting (SFO) strictly ○?

It is not just that naive erasure fails; the operation mathematically forces an **exponential blow-up** in the representation size.

- Logically, forgetting a variable x is defined as:
$$\exists x. \Sigma \equiv (\Sigma \mid x) \vee (\Sigma \mid \neg x).$$
- Conditioning (**CD**) is ✓, so $(\Sigma \mid x)$ and $(\Sigma \mid \neg x)$ are both valid, compact ODNFs.
- **The Proof:** We must combine them with an OR ($\vee C$). Making two arbitrary ODNFs orthogonal to each other requires resolving all their intersections. In the worst case, this shatters the terms and causes an **exponential explosion**. Thus, no poly-time algorithm can exist.

Succinctness: Where does ODNF fit?

- ODNF sits perfectly between the “too relaxed” DNF and the “too strict” MODS.



- $\text{DNF} <_s \text{ODNF}$: Making terms orthogonal can require an exponential blow-up in the number of terms.
- $\text{ODNF} <_s \text{MODS}$: A single ODNF term of size k compactly represents 2^{n-k} models, avoiding the need to list them all.

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Taking Advantage of the Solver's Trace

- When a SAT solver evaluates a CNF instance Σ , it systematically explores the search space of interpretations until it finds a model (or proves unsatisfiability).
- **The KC Insight:** We can use this *exact same search space* to compile Σ . We just prevent the solver from stopping when it finds the first model.
- By recording the trace of the solver's decisions, we generate a compiled structural form of the formula.

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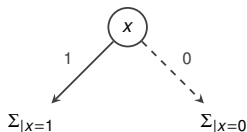
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The Decision Tree (DT)

- **Shannon Expansion:** A formula Σ can be split on any variable x :

$$\Sigma \equiv (x \wedge \Sigma_{|x=1}) \vee (\neg x \wedge \Sigma_{|x=0})$$



- A Decision Tree is built by recursively applying Shannon Expansion.
- **Logically:** It is the joined representation of a deterministic DNF for Σ and a deterministic DNF for $\neg\Sigma$.

Decision Tree Compiler

Formula

$$\Sigma = (\neg x \vee y \vee \neg z) \wedge (x \vee y \vee z) \wedge (y \vee t \vee u) \wedge (\neg y \vee t \vee u)$$

Decision Tree Compiler

Formula

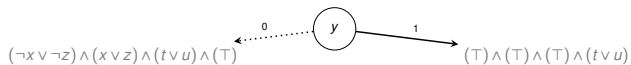
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Decision Tree Compiler

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Decision Tree Compiler

Formula

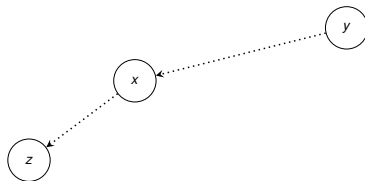
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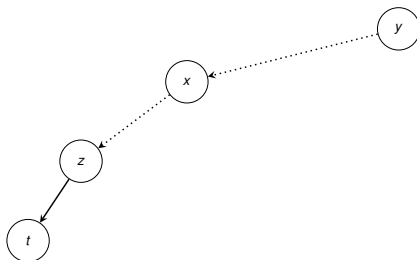
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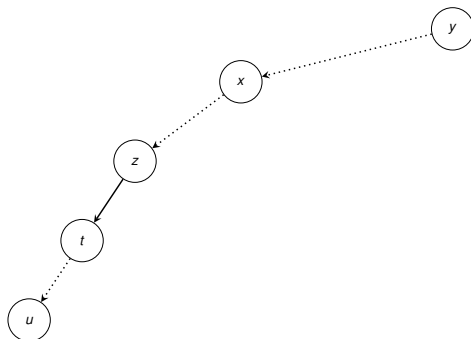
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Decision Tree Compiler

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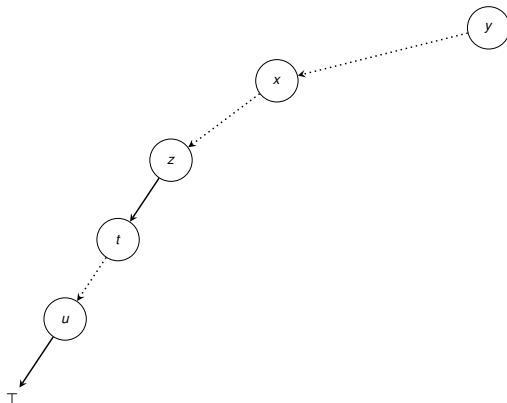
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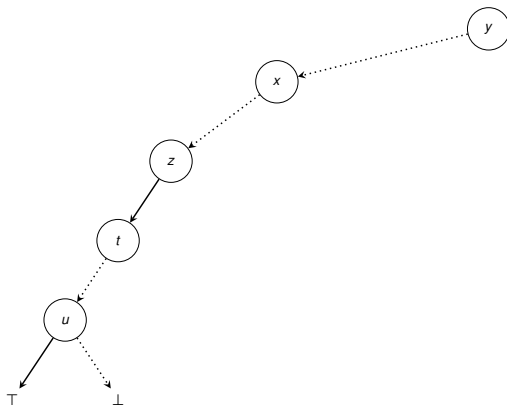
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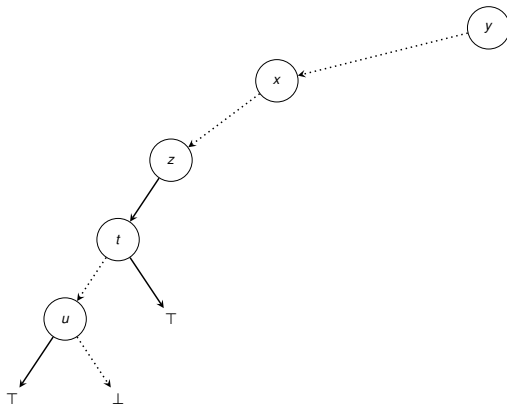
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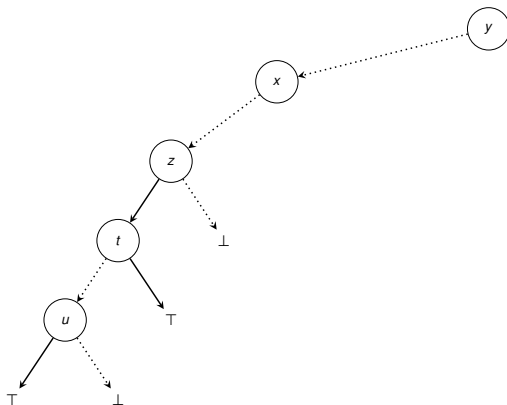
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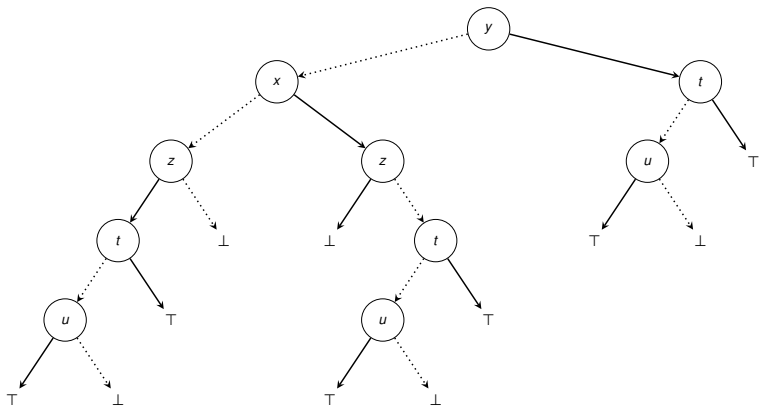
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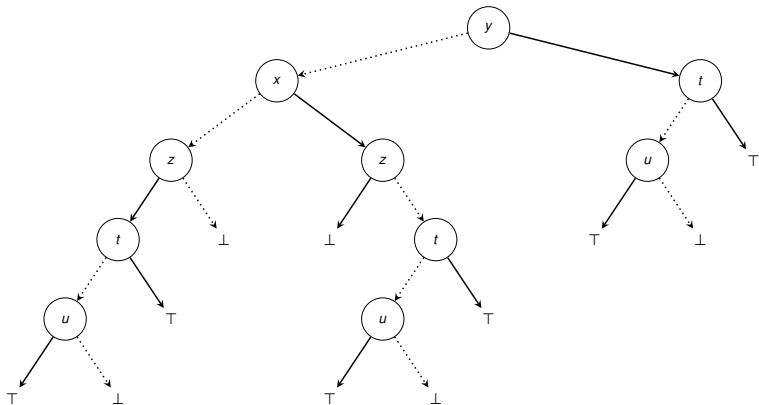
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$$\Sigma = (\neg x \vee y \vee \neg z) \wedge (x \vee y \vee z) \wedge (y \vee t \vee u) \wedge (\neg y \vee t \vee u)$$



- **Problem:** The tree size is exponential in the number of variables (2^n). We re-solve similar sub-problems (like $t \vee u$) multiple times.

The KC Map for DT: Queries

$\mathcal{L}$	CO	VA	CE	IM	EQ	SE	CT	ME
Circ	○	○	○	○	○	○	○	○
CNF	○	✓	○	✓	○	○	○	○
DNF	✓	○	✓	○	○	○	○	✓
ODNF	✓	✓	✓	✓	✓	✓	✓	✓
MODS	✓	✓	✓	✓	✓	✓	✓	✓
DT	✓	✓	✓	✓	✓	✓	✓	✓

- **A Powerful Result:** Decision Trees (DT) satisfy **all** the queries offered by Ordered Binary Decision Diagrams (OBDD_<).
- **Why?** Because DT is structurally a tree, computing the intersection of two trees T_1, T_2 yields a size bounded by $\mathcal{O}(|T_1| \times |T_2|)$. This makes complex binary queries like Equivalence (**EQ**) and Sentential Entailment (**SE**) strictly polynomial!

The KC Map for DT: Transformations

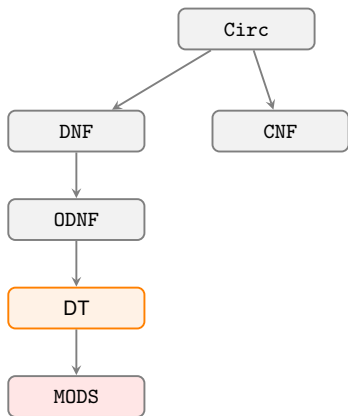
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Circ	✓	○	✓	✓	✓	✓	✓	✓
CNF	✓	○	✓	✓	✓	○	✓	○
DNF	✓	✓	✓	○	✓	✓	✓	○
ODNF	✓	○	○	○	✓	○	○	○
MODS	✓	✓	✓	○	✓	○	✓	○
DT	✓	○	✓	○	✓	○	✓	✓

■ Inheriting OBDD_< Traits:

- **Negation ($\neg C$):** Trivial ✓. Just swap the $\top$ and $\perp$ leaves.
- **Bounded Conjunction/Disjunction ($\wedge BC, \vee BC$):** ✓, because the product of a *constant number* of trees remains polynomial.
- **Singleton Forgetting (SFO):** ✓. Existential quantification reduces to applying an OR operation on the branches, which is poly-time due to $\vee BC$.

- **Weakness:** Unbounded operations (**FO**, $\wedge C$, $\vee C$) remain ○.

Succinctness: The Problem with Trees



Why is $ODNF <_s DT$?

An ODNF allows terms of different lengths to coexist natively. A DT forces rigid, nested branching on variables, which can require exponentially more space to represent the exact same orthogonal logic.

The Tree Bottleneck

While DT has incredible query power (matching $OBDD_{<}$), it lacks a DAG structure. **Isomorphic subgraphs cannot be shared**; they must be duplicated in memory. This lack of compression is why we must transition to DAG-based languages!

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The Achilles Heel of DT: The Parity Function

- Consider the Boolean Parity function (which evaluates to True if an even number of variables are True):

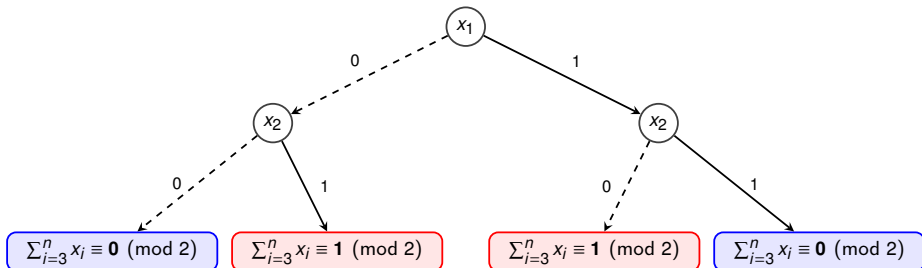
$$\sum_{i=1}^n x_i \equiv 0 \pmod{2}$$

Why does DT fail here?

- To determine parity, we **must** read the value of every single variable x_i . We can never short-circuit the evaluation.
- In a strict Decision Tree, this forces us to build a complete binary tree of depth n .
- **The Consequence:** The DT will always have $\mathcal{O}(2^n)$ nodes. It effectively degenerates into explicitly listing all interpretations (like MODS)!

The Solution: Caching (Sub-circuit Sharing)

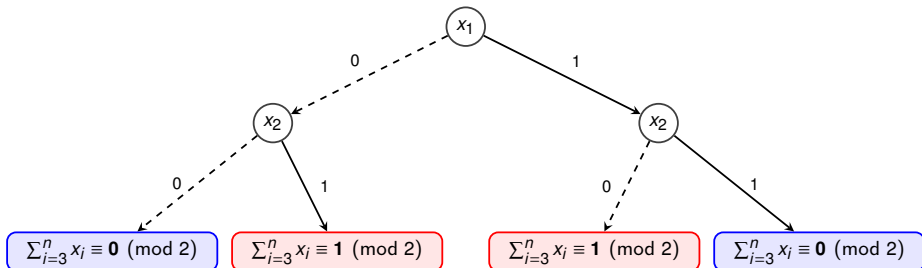
- Let's trace the Shannon expansion of the parity function after two variables.



- The Observation:** Despite having 4 branches, there are only **two** unique sub-problems remaining!

The Solution: Caching (Sub-circuit Sharing)

- Let's trace the Shannon expansion of the parity function after two variables.



- The Observation:** Despite having 4 branches, there are only **two** unique sub-problems remaining!
- The DAG Trick:** By caching and merging isomorphic nodes, the size of this representation drops from $\mathcal{O}(2^n)$ down to just $\mathcal{O}(n)$!

Free Binary Decision Diagram (FBDD)

Formula

$$\Sigma = (\neg x \vee y \vee \neg z) \wedge (x \vee y \vee z) \wedge (y \vee t \vee u) \wedge (\neg y \vee t \vee u)$$

Free Binary Decision Diagram (FBDD)

Formula

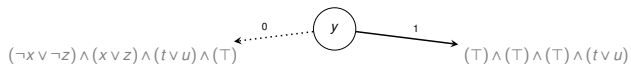
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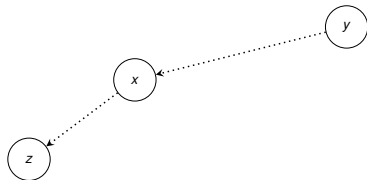
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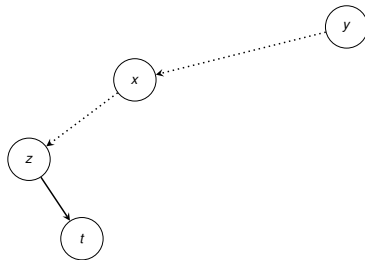
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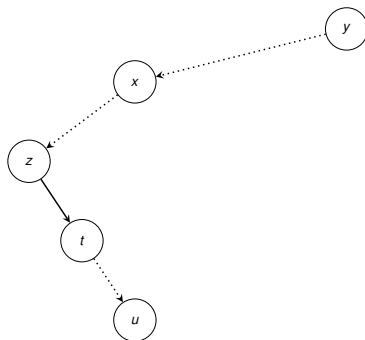
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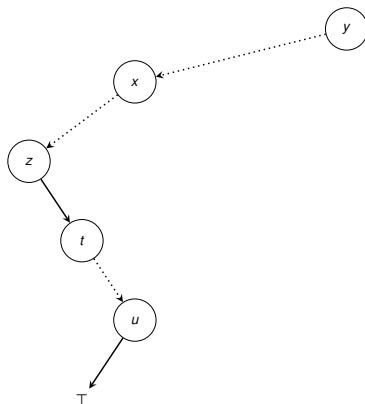
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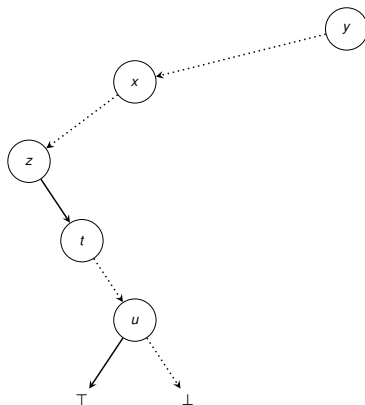
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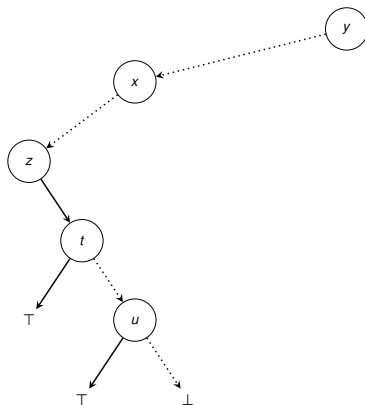
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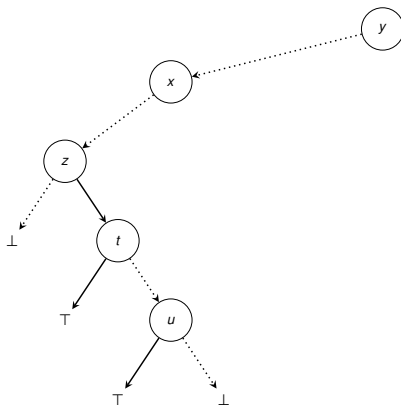
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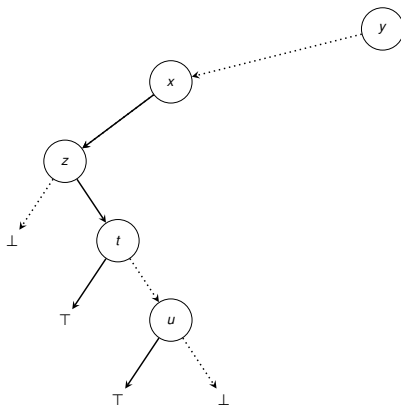
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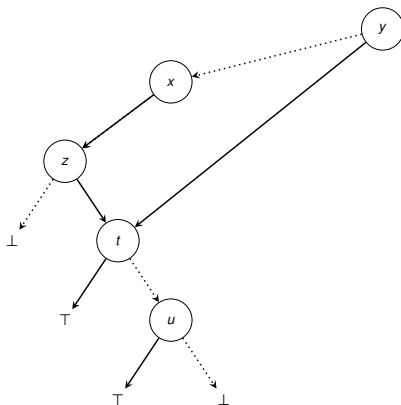
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$$\Sigma = (\neg x \vee y \vee \neg z) \wedge (x \vee y \vee z) \wedge (y \vee t \vee u) \wedge (\neg y \vee t \vee u)$$



The KC Map for FBDD: Queries

$\mathcal{L}$	CO	VA	CE	IM	EQ	SE	CT	ME
Circ	○	○	○	○	○	○	○	○
CNF	○	✓	○	✓	○	○	○	○
DNF	✓	○	✓	○	○	○	○	✓
ODNF	✓	✓	✓	✓	✓	✓	✓	✓
MODS	✓	✓	✓	✓	✓	✓	✓	✓
DT	✓	✓	✓	✓	✓	✓	✓	✓
FBDD	✓	✓	✓	✓	○	○	✓	✓

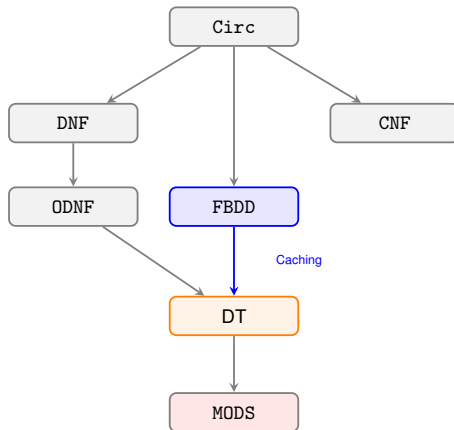
- **The Trade-off for Compression:** By allowing the tree to merge into a DAG (FBDD), we successfully shrink the size while preserving poly-time Model Counting (**CT**) and Consistency (**CO**).
- **What we lose:** Because the paths can now merge and variables can be tested in different orders on different branches, Equivalence (**EQ**) and Sentential Entailment (**SE**) become **NP-hard**.

The KC Map for DT: Transformations

$\mathcal{L}$	CD	FO	SFO	$\wedge C$	$\wedge BC$	$\vee C$	$\vee BC$	$\neg C$
Circ	✓	○	✓	✓	✓	✓	✓	✓
CNF	✓	○	✓	✓	✓	○	✓	○
DNF	✓	✓	✓	○	✓	✓	✓	○
ODNF	✓	○	○	○	✓	○	○	○
MODS	✓	✓	✓	○	✓	○	✓	○
DT	✓	○	✓	○	✓	○	✓	✓
FBDD	✓	○	○	○	○	○	○	✓

- **Negation remains easy:** $\neg C$ is still ✓ because we can simply swap the $\top$ and $\perp$ leaves of the DAG.
- **Binary operations fail:** Merging two DAGs that have completely different, unrestricted variable orderings causes an exponential blow-up. Consequently, we lose bounded Transformations.

Succinctness: DAGs vs Trees



- $\text{FBDD} <_s \text{DT}$: By allowing isomorphic subgraphs to be shared, FBDD is **strictly more succinct** than DT.
- *Recall the Parity function*: It requires $\mathcal{O}(2^n)$ size in DT, but only $\mathcal{O}(n)$ size in FBDD!

Building the FBDD Compiler: Memoization

The Naive Approach

Compiling a full DT first and then searching for identical sub-trees to merge them is impractical (the DT would not fit in memory!).

The Solution: On-the-fly Caching

- We maintain a **Cache** (a hash map) storing pairs of $\langle \text{CNF}, \text{FBDD} \rangle$.
- At every new decision node in the solver, we check if the current remaining CNF sub-problem is already in the map.
- **If yes:** We link to the existing FBDD node (sharing).
- **If no:** We compile it, and add the result to the cache.

The Implementation Catch

- Testing if two formulas are *logically equivalent* is coNP-complete. We cannot do this at every step!
- **In practice:** We check for **Syntactic Identity** (are the clauses exactly the same, up to ordering?). This is fast via hashing.

Component Caching

Component Caching: The Concept

What is Component Caching?

In model counting, we often encounter the same sub-problems (components) in different branches of the search tree. Caching stores these results to avoid re-solving them (Sang et al., 2004; Lagniez and Marquis, 2021).

- **The Caching Scheme** (r_c): A mapping that converts a sub-formula φ into a **Cache Key** (signature).

$$\varphi \xrightarrow{r_c} \text{Key}$$

- **The Cache:** A hash map storing: $\text{Key} \mapsto \text{Circ.}$
- **Correctness Condition:** If two sub-formulas generate the same key, they must be logically equivalent.

$$r_c(\varphi_1) = r_c(\varphi_2) \implies \varphi_1 \equiv \varphi_2$$

Step 1: Representing a Single Clause

Since a component is a set of clauses, we first need a way to represent individual clauses in the cache key.

1. Literal Representation

The clause is explicitly stored as a sorted list of literals.

- **Format:** Integers (Variable IDs).
- **Sign:** Positive ID for x_i , negative for $\neg x_i$.
- **Terminator:** Ends with 0.

Ex: $(x_1 \vee \neg x_3) \rightarrow [1, -3, 0]$

2. Index Representation

The clause is represented by its unique ID from the original CNF.

- **Format:** A single integer (Index) and a list of the variables indices.
- **Pros:** Very compact (low memory).
- **Cons:** Less precise.

Ex: #42: $(x_1 \vee x_3) \rightarrow [42][1, 3]$

Step 2: Representing the Formula (The Scheme)

Which clauses should be included in the Cache Key?

■ **Scheme A: “Standard” All Clauses (cachet) (Sang et al., 2004)**

- Include **all** clauses belonging to the current component.
- *Safe but verbose.*
- **Requirement:**
 - *Literal Rep:* Variables are implicit (contained in the literals).
 - *Index Rep:* Need to explicitly store the variable set.

■ **Scheme B: “Non-Binary” (sharpSAT) (Muisse et al., 2010)**

- Include only clauses of **size** > 2 .
- *Rationale:* Binary clauses are numerous but imply little structure. Excluding them saves memory and increases cache hits (abstraction).
- **Requirement:** Need to explicitly store which variables are involved.

■ **Scheme C: “Not Touched” (D4) (Lagniez and Marquis, 2017a)**

- Include only clauses that have been **modified** (shortened) by the current partial assignment.
- *Rationale:* If a clause is “untouched” (original state), it is implicitly defined by the set of variables involved in the component.
- **Requirement:** Need to explicitly store which variables are involved.

Running Example: Setup

Let us consider a specific CNF Σ and a partial assignment.

Original CNF Σ

Indices:

$$1 \quad (x_1 \vee x_2 \vee x_4)$$

$$2 \quad (x_2 \vee x_3 \vee x_4)$$

$$3 \quad (x_3 \vee x_4 \vee x_5)$$

$$4 \quad (x_1 \vee \neg x_6)$$

Current State (φ)

Decision: Set $x_2 = \text{T}$. **BCP (Unit Propagation):**

- Clause 1 ($x_1 \vee \text{T} \vee x_4$) is **Satisfied** (Removed).
- Clause 2 ($\text{T} \vee x_3 \vee x_4$) is **Satisfied** (Removed).

Remaining Component φ :

$$(x_3 \vee x_4 \vee x_5) \wedge (x_1 \vee \neg x_6)$$

Active Variables: $\{1, 3, 4, 5, 6\}$

Example 1: “All Clauses” Scheme

Stores the variable set plus **every** active clause.

$$\text{Remaining Clauses: } \underbrace{(x_3 \vee x_4 \vee x_5)}_{\text{Clause 3}} \wedge \underbrace{(x_1 \vee \neg x_6)}_{\text{Clause 4}}$$

Representations

- **Literal Representation:** Explicitly list literals for both clauses.

$$r(\varphi) = ([\underbrace{3, 4, 5, 0}_{C_3}, \underbrace{1, -6, 0}_{C_4}])$$

- **Index Representation:** List the IDs of the active clauses plus variables.

$$r(\varphi) = ([1, 3, 4, 5, 6], \quad [3, 4])$$

Example 2: “Non-Binary” Scheme (sharpSAT)

Stores variables plus only **large** clauses (Size > 2).

$$\text{Active Clauses: } \underbrace{(x_3 \vee x_4 \vee x_5)}_{\text{Size 3 (Keep)}} \wedge \underbrace{(x_1 \vee \neg x_6)}_{\text{Size 2 (Drop)}}$$

Representations

- **Literal Representation:** Only the ternary clause is stored explicitly.

$$r(\varphi) = ([1, 3, 4, 5, 6], \quad [3, 4, 5, 0])$$

- **Index Representation:** Only the index of the ternary clause.

$$r(\varphi) = ([1, 3, 4, 5, 6], \quad [3])$$

Example 3: “Not Touched” Scheme (D4)

Stores variables plus only **modified** clauses.

■ $C_3 : (x_3 \vee x_4 \vee x_5)$. Original: Same. **Status: Untouched.**

■ $C_4 : (x_1 \vee \neg x_6)$. Original: Same. **Status: Untouched.**

Since neither clause was modified by the assignment $x_2 = \top$ (they didn't contain x_2), they are implicit.

Representations

■ **Literal & Index Representation:** The clause list is empty! The component is identified solely by its variables.

$$r(\varphi) = ([1, 3, 4, 5, 6], [\emptyset])$$

Note: If a clause had been shortened (e.g., $(x_3 \vee x_4)$ from an original $(x_3 \vee x_4 \vee \neg x_2)$), it would be included here.

Cache Management: The Cleaning Problem

The Problem

The cache grows indefinitely.

- Storing millions of components consumes gigabytes of RAM.
- Hash collisions increase, slowing down lookups.

The Strategy

We need a **Cleaning Policy** to periodically remove “low-value” entries.

$$\text{Value}(\textit{Entry}) = f(\text{Age}, \text{Activity}, \text{Size})$$

Cache Cleaning Strategies (I)

1. Cachet (Age-Based)

Old entries are useless.

- **Metric:** Age (time since creation).
- **Trigger:** When cache is full (e.g., 2^{21} entries).
- **Action:** Delete the oldest entries (keep 25% newest).

2. SharpSAT (Activity-Based)

Used entries are useful.

- **Metric:** Score (Activity).
 - Init: Age.
 - Hit: Reset score (boost).
 - Periodic: Decay all scores.
- **Trigger:** Cache exceeds max size fraction.
- **Action:** Delete lowest scores.

Cache Cleaning Strategies (II)

3. D4 (Size-Aware)

Small components (few variables) are hit more often than large ones. A “one-size-fits-all” score is inefficient.

- 1 Track Performance by Size:** Compute the hit ratio for each component size s :

$$r(s) = \frac{\text{Hits on components of size } s}{\text{Total components of size } s}$$

- 2 Identify “Dead” Entries:** An entry e is flushed if:

- Its size class is performing poorly ($r(|e|) < \text{Threshold}$).
- AND it hasn't been used recently ($\text{flag}(e) = \text{false}$).
- AND its activity score is zero.

- 3 Aging:** For kept entries, divide score by 2. If score drops to 0, uncheck flag.

The Limit of FBDD: Independent Components

- How efficiently can we compile the following formula into an FBDD?

$$\Sigma = \bigwedge_{i=1}^n \left(x_1^i \vee x_2^i \vee \dots \vee x_n^i \right)$$

The Problem

- Each clause is completely independent (they share no variables).
- An FBDD **forces** us to make sequential decision branches, weaving these independent problems together into a single path.
- This inability to cleanly separate independent sub-problems causes an unavoidable exponential blow-up in the DAG size!

Outline

1 How to model using SAT

2 Knowledge Compilation

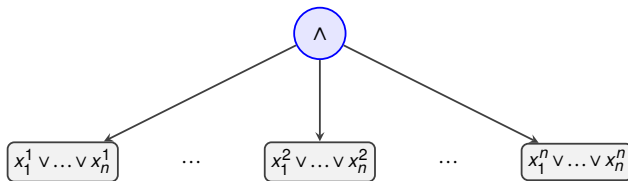
- Knowledge Compilation Map (Darwiche and Marquis, 2002)
- SAT Based Compilers
- **Top-Down Compilers**
 - DT
 - FBDD
 - **decision-DNNF**
 - AFF
 - ADT
 - EADT
- Dynamic Programming and Compilation
- Bottom-Up Compilers

3 d-DNNF Queries

4 Conclusion and References

The Solution: Decomposition

- We observe that the clauses share no variables. They are disjoint.
- Can we compile the clauses separately and aggregate them using an explicit **AND** ($\wedge$) node while preserving key queries like counting?
- **Yes!** Because the variables are disjoint, the number of models of the AND node is exactly the **product** of the models of its children.



KC Map for decision-DNNF: Queries

$\mathcal{L}$	CO	VA	CE	IM	EQ	SE	CT	ME
Circ	○	○	○	○	○	○	○	○
CNF	○	✓	○	✓	○	○	○	○
DNF	✓	○	✓	○	○	○	○	✓
ODNF	✓	✓	✓	✓	✓	✓	✓	✓
MODS	✓	✓	✓	✓	✓	✓	✓	✓
DT	✓	✓	✓	✓	✓	✓	✓	✓
FBDD	✓	✓	✓	✓	○	○	✓	✓
decision-DNNF	✓	✓	✓	✓	?	○	✓	✓

- **No Loss in Query Power!** Adding Decomposable AND nodes preserves all the poly-time queries we had in FBDD, including Model Counting (**CT**).
- *Note:* Whether Equivalence (**EQ**) is poly-time for decision-DNNF remains an open question (?) in the strict Darwiche & Marquis framework.

KC Map for decision-DNNF: Transformations

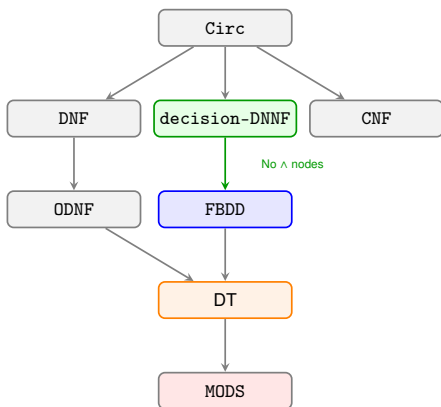
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FBDD	✓	○	○	○	○	○	○	✓
decision-DNNF	✓	○	○	○	○	○	○	?

The Loss of Negation ($\neg C$)

For DT and FBDD, negation was simply swapping $\top$ and $\perp$ leaves. But decision-DNNF introduces explicit AND ($\wedge$) nodes!

- By De Morgan's laws, negating an AND node creates an OR node ($\vee$).
- We have **no guarantee** that the resulting OR node is deterministic (orthogonal). If determinism breaks, we can no longer count models!

Succinctness: The Power of AND Nodes



- $\text{decision-DNNF} <_s \text{FBDD}$: By allowing independent sub-formulas to be explicitly split via AND nodes, we prevent them from being needlessly interleaved.
- **The Result:** decision-DNNF is **strictly more succinct** than FBDD. Our disjoint CNF example requires exponential size in FBDD, but only **linear size** in decision-DNNF!

Decision-DNNF (decision-DNNF)

Formula

$$\Sigma = (\neg x \vee y \vee \neg z) \wedge (x \vee y \vee z) \wedge (y \vee t \vee u) \wedge (\neg y \vee t \vee u)$$

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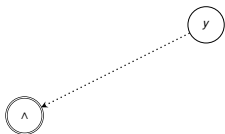
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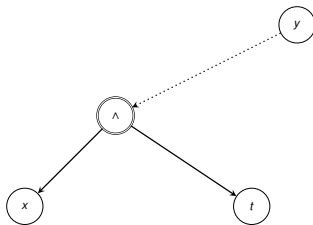
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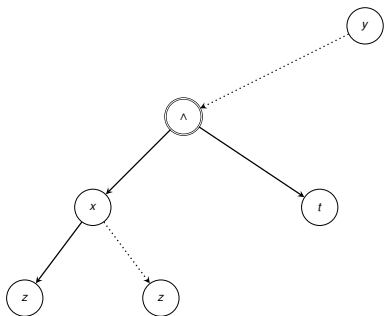
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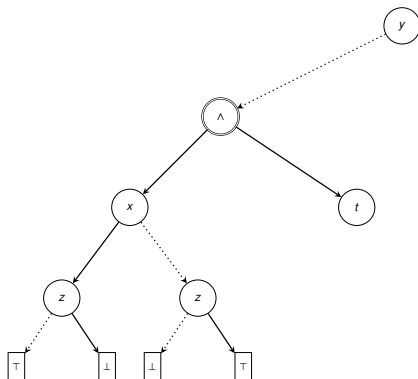
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Formula

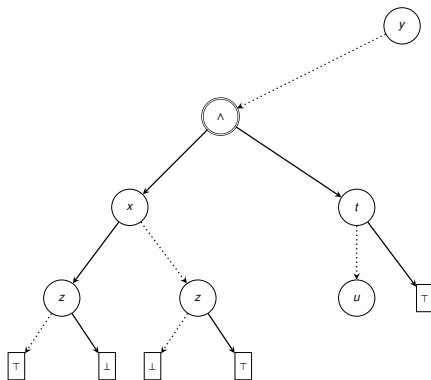
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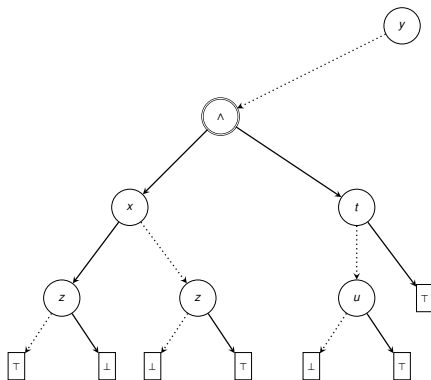
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Decision-DNNF (decision-DNNF)

Formula

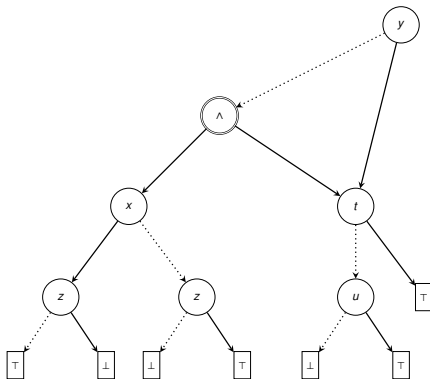
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Decision-DNNF (decision-DNNF)

Formula

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Branching Heuristics

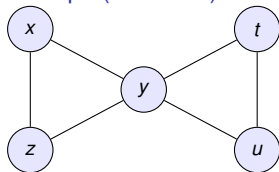
Graph Representations of CNF

We represent the CNF Σ as a graph to find structure.

Formula with 4 Clauses

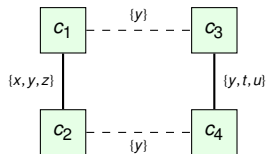
$$\Sigma = \underbrace{(\neg x \vee y \vee \neg z)}_{c_1} \wedge \underbrace{(x \vee y \vee z)}_{c_2} \wedge \underbrace{(y \vee t \vee u)}_{c_3} \wedge \underbrace{(\neg y \vee t \vee u)}_{c_4}$$

1. Primal Graph (Variables)



Shows variables interacting. Note that c_3 and c_4 cover the same clique $\{y, t, u\}$.

2. Dual Graph (Clauses)

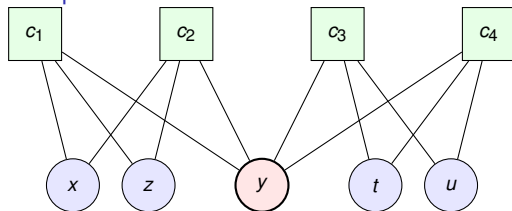


Shows clusters of clauses. Cutting y clearly separates $\{c_1, c_2\}$ from $\{c_3, c_4\}$.

Graph Representations: The Incidence Graph

- The *Incidence Graph* is a **Bipartite Graph** that preserves the exact structure of the formula without adding “clique edges” (unlike the Primal Graph).
- **Nodes:** $V \cup C$ (Variables and Clauses).
- **Edges:** Edge (v, c) exists if variable v appears in clause c .

Example: Incidence Graph of Σ



- **Visual Insight:** Notice how variable y acts as the sole *cut vertex* (bridge).
- If y is assigned (removed), the graph splits into two disjoint connected components: $\{c_1, c_2\}$ and $\{c_3, c_4\}$.

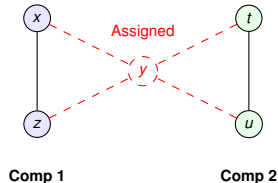
Computing Decomposition: Algorithms

How do we identify independent components?

Dynamic Detection: Component Retrieval

During search (at each decision node), the graph changes.

- **DFS / BFS:** Standard traversal.
- **Union-Find:** Maintains disjoint sets.



The Bottleneck: Cost vs. Gain

Computing connected components has polynomial complexity $O(V + E)$, but it must be done at **every decision node**.

- In a search tree with millions of nodes, this overhead is significant.
- Standard graph libraries are too slow; solvers use specialized, lightweight adjacency lists.

Why do we pay this price?

Despite the cost, component detection is the **single most important feature** for tractable counting.

The Complexity Gap:

- **Without Decomposition:** Complexity is $O(2^n)$.
- **With Decomposition:** If the graph splits into two roughly equal parts, complexity drops to $O(2^{n/2} + 2^{n/2}) \approx O(2^{n/2})$.

The exponential reduction in search space drastically outweighs the polynomial cost of the BFS.

Classical SAT Heuristics

- **Goal:** Pick the variable most likely to lead to a conflict (to learn clauses fast) or to satisfy the formula.

1. Structure Aware

- **MOMS:** Maximum Occurrence in clauses of Minimum Size (Crawford and Auton, 1996).
- **Jeroslow-Wang:** Weighted occurrence (favors short clauses exponentially) (Jeroslow and Wang, 1990).

2. Activity-Based

VSIDS (Variable State Independent Decaying Sum) (Moskewicz et al., 2001)

- **Mechanism:** Increment a counter for a variable every time it participates in conflict analysis. Decay all counters periodically.
- **Effect:** Focuses search on the “hot” part of the formula (recent conflicts).

Adapting to Counting: VSADS

The Counting Dilemma

- **SAT solvers** want conflicts (to prove UNSAT).
- **Counters** want **decomposition** (to multiply sub-counts).
- Pure VSIDS might stick to one giant component for too long.

VSADS (Sang et al., 2004)

Variable State Aware Decaying Sum

A hybrid score combining activity and structural relevance:

$$\text{Score}(v) = \alpha \cdot \mathbf{Activity}(v) + \beta \cdot \mathbf{Count}(v)$$

- **Activity:** Standard VSIDS score (conflicts).
- **Count:** How many times v appears in the *current* formula.
- **Dynamic:** If many conflicts occur, α dominates (act like CDCL). If formula simplifies, β dominates (act like MOMS/Decomposition).

Cache-Aware Branching: CSVSADS

- **Observation:** In model counting, **Component Caching** is the primary speedup mechanism.
- We should branch on variables that define *cache keys*, increasing the probability of a **Cache Hit**.

CSVSADS (Sharma et al., 2019)

Cache Score based VSADS

$$\text{Score}(v) = \text{VSIDS}(v) + \mathbf{CacheScore}(v)$$

How **CacheScore** works:

- When storing a component in the cache, increase score of variables appearing in that component.
- **Intuition:** If a variable appears in many cached components, assigning it might lead to a state we have seen before.
- Reduces the number of unique components generated during search.

Structure-Guided Branching: The Static Approach (C2D)

The Idea (Darwiche, 2004)

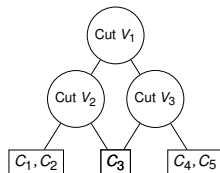
Instead of letting the solver wander (VSIDS), we force it to follow a **pre-computed roadmap** that guarantees decomposition.

- 1 **Offline Phase:** Construct a **dtree** (Decomposition Tree) for the CNF.

- A binary tree where leaves are clauses.
- Internal nodes represent the union of clauses.

- 2 **Online Phase (Compilation):**

- Visit the dtree nodes.
- Branch on the **cutset** variables (variables shared between the left and right child).



Variables in "Cut" are instantiated first to separate the clauses below.

- **Pros:** Guarantees low treewidth behavior.
- **Cons: Static.** Does not adapt to the simplifications (unit propagations) occurring during the search.

Dynamic Structural Branching (D4)

The idea (Lagniez and Marquis, 2017a)

D4 integrates graph partitioning directly into the search loop:

- 1 **Compute Cutset:** Call a partitioner (e.g., KaHyPar) to find a separator S for the current component.
- 2 **Filter:** Restrict branching candidates to S (ignoring others).
- 3 **Select:** Choose the best variable $v \in S$ using VSADS.
- 4 **Repeat:** When S is exhausted, re-partition the remaining graph.

Advantage

Computes cuts on the **simplified** graph (post-propagation). Often yields much smaller separators than static methods (C2D).

Drawbacks

- **Overhead:** Partitioning at runtime is expensive.
- **“Tunnel Vision”:** Ignores high-activity variables outside S , potentially delaying conflict resolution.

Structure-Guided Branching (sharpSAT-TD)

Concept: Hybrid Guidance (Korhonen and Järvisalo, 2021)

Theory suggests counting is efficient for small treewidth k ($O(2^k n)$), but practical k is often too large. **sharpSAT-TD** uses Tree Decomposition (TD) as a **soft guide** rather than a rigid order.

Modified Heuristic

$$\text{sc}(x) = \underbrace{\text{act}(x) + \text{fr}(x)}_{\text{VSADS}} - \underbrace{C \cdot \min_{t \in T_x} d_T(t)}_{\text{Struct. Penalty}}$$

Key Components:

- **Penalty:** $d_T(t) \in [0, 1]$ is the distance of bag t from the root.
 - **Effect:** Variables at the “top” of the TD (separators) have lower penalty $\rightarrow$ higher priority.
-
- **Tuning:** C scales with width w .
 - **Large C :** Strictly follows decomposition order.
 - **Small C :** Behaves like standard conflict-driven search.

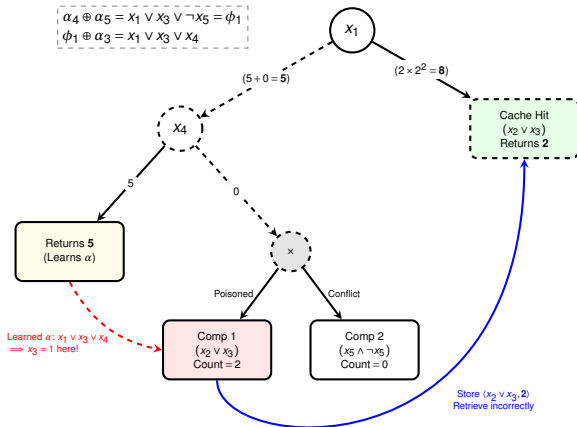
Cache Inconsistency

Cache Inconsistency: Learned Clauses

Input Formula & Scenario

$$\Sigma = \{ \underbrace{(\neg x_1 \vee x_2 \vee x_3)}_{\alpha_1}, \underbrace{(x_1 \vee x_2 \vee x_3)}_{\alpha_2}, \underbrace{(x_1 \vee x_4 \vee x_5)}_{\alpha_3}, \underbrace{(x_1 \vee x_4 \vee \neg x_5)}_{\alpha_4}, \underbrace{(x_1 \vee x_3 \vee \neg x_4 \vee \neg x_5)}_{\alpha_5} \}$$

Total: $5 + 8 = 13 \neq 17$ (Error)



Cache Inconsistency: Handling Conflicts

The Problem: Context Dependence

- CDCL learned clauses (α) are globally valid but may not be implied by the local component.
- If α triggers propagations inside a component, the computed count is **smaller** than the true count (over-constrained).

The Consequence

Cache Poisoning

- In the example, we stored $\langle A, 2 \rangle$.
- This value was only true because the branch eventually led to a conflict (UNSAT).
- If kept, it corrupts future valid branches.

The Solution

Reactive Cache Cleaning

- **Track:** Keep track of entries added during the exploration of the current sub-tree.
- **Trigger:** If the sub-tree returns **0 (UNSAT)**.
- **Action: Remove** all entries added during this exploration.
- **Result:** The “poisoned” entry $\langle A, 2 \rangle$ is deleted because the branch ‘x4=0’ failed.

Outline

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- Knowledge Compilation Map (Darwiche and Marquis, 2002)
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- **Top-Down Compilers**
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3 d-DNNF Queries

4 Conclusion and References

■ Objective:

- Define new propositional languages (ADT, EADT) that offer polynomial-time Model Counting (**CT**).
- Place them accurately on the Knowledge Compilation map.

■ ADT and EADT are built by combining **Affine Formulae** (AFF) and **Decision Trees** (DT):

- AFF satisfies **CT** natively but is logically *incomplete* (it cannot express all Boolean functions).
- *decision-DNNF* and its subsets (FBDD, DT) are *complete* and satisfy **CT**.

■ The Core Idea: Can we merge the algebraic counting power of AFF with the structural completeness of DT?

Affine Formulae (AFF)

- An **affine clause** is a finite XOR-disjunction of literals (including $\perp$ and $\top$).
- A **simplified affine clause** is a finite XOR-disjunction of pure atoms, plus at most one occurrence of $\top$.
- **Property:** Every affine clause can be rewritten in *linear time* into an equivalent simplified affine clause:

$$x \oplus \neg y \oplus \neg z \mapsto x \oplus y \oplus z$$

- An **affine formula** (Φ) is a finite conjunction of affine clauses:

$$\Phi = (x \oplus \neg y \oplus \neg z) \wedge (x \oplus u \oplus \neg w) \wedge (w \oplus y \oplus \neg z)$$

Model Counting for AFF

The CT Query

Model counting can be achieved in **polynomial time** when the input is an Affine Formula.

- We do this by turning Φ into reduced row echelon form using the **Gauss Elimination** algorithm (in the field of characteristic 2).

$$\Phi = \left(\begin{array}{cccc|c} x & y & z & & 1 \\ x & & & u & 0 \\ & y & z & w & 0 \end{array} \right) \begin{array}{l} (E1) \\ (E2) \\ (E3) \end{array}$$

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- The resulting system of linear equations has $\#f = 2$ “free” variables (w and z). Therefore, the model count is exactly $\|\Phi\| = 2^{\#f} = 4$.

KC Map for AFF: Queries

$\mathcal{L}$	CO	VA	CE	IM	EQ	SE	CT	ME
Circ	○	○	○	○	○	○	○	○
CNF	○	✓	○	✓	○	○	○	○
DNF	✓	○	✓	○	○	○	○	✓
ODNF	✓	✓	✓	✓	✓	✓	✓	✓
MODS	✓	✓	✓	✓	✓	✓	✓	✓
DT	✓	✓	✓	✓	✓	✓	✓	✓
FBDD	✓	✓	✓	✓	○	○	✓	✓
decision-DNNF	✓	✓	✓	✓	?	○	✓	✓
AFF	✓	✓	✓	✓	✓	✓	✓	✓

- **A Perfect Score:** AFF satisfies every single query in polynomial time, matching DT and MODS!

KC Map for AFF: Transformations

$\mathcal{L}$	CD	FO	SFO	$\wedge \mathbf{C}$	$\wedge \mathbf{BC}$	$\vee \mathbf{C}$	$\vee \mathbf{BC}$	$\neg \mathbf{C}$
Circ	✓	○	✓	✓	✓	✓	✓	✓
CNF	✓	○	✓	✓	✓	○	✓	○
DNF	✓	✓	✓	○	✓	✓	✓	○
ODNF	✓	○	○	○	✓	○	○	○
MODS	✓	✓	✓	○	✓	○	✓	○
DT	✓	○	✓	○	✓	○	✓	✓
FBDD	✓	○	○	○	○	○	○	✓
decision-DNNF	✓	○	○	○	○	○	○	?
AFF	✓	✓	✓	✓	✓	!	!	!

- **The Incompleteness Penalty:** AFF cannot express OR ($\vee$) natively. Therefore, operations like Disjunction ($\vee \mathbf{C}, \vee \mathbf{BC}$) and Negation ($\neg \mathbf{C}$) are not guaranteed to produce a valid Affine formula.

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Affine Decision Tree (ADT)

- **The Idea:** Generalize the standard Shannon expansion by replacing the literal (x) with an **Affine Clause** (δ).

Generalized Shannon Expansion

$$\Phi \equiv ((x \Leftrightarrow \neg\delta) \wedge \Phi_{|x \leftarrow \neg\delta}) \vee ((x \Leftrightarrow \delta) \wedge \Phi_{|x \leftarrow \delta})$$

- Since $(x \Leftrightarrow \neg\delta) \equiv x \oplus \delta$ and $(x \Leftrightarrow \delta) \equiv \neg x \oplus \delta$, we rewrite it as:

$$\Phi \equiv ((x \oplus \delta) \wedge \Phi_{|x \leftarrow \delta \oplus \top}) \vee ((\neg x \oplus \delta) \wedge \Phi_{|x \leftarrow \delta})$$

Generation and Model Counting

- Let's compile $\Phi = (\neg x \vee y \vee \neg z) \wedge (x \vee y \vee z) \wedge (y \vee t \vee u)$

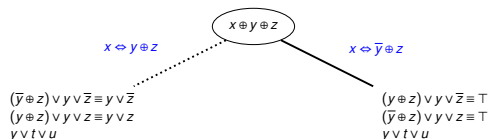
Generation and Model Counting

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$$x \oplus y \oplus z$$

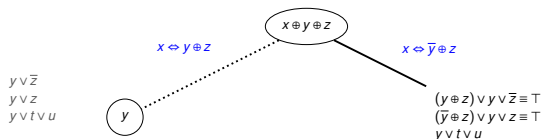
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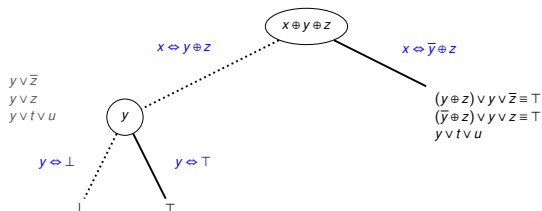
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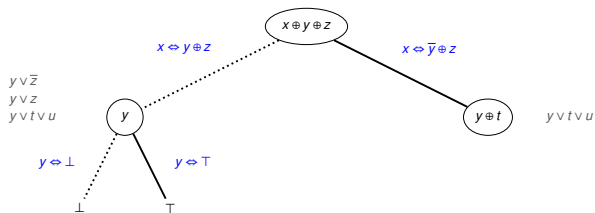
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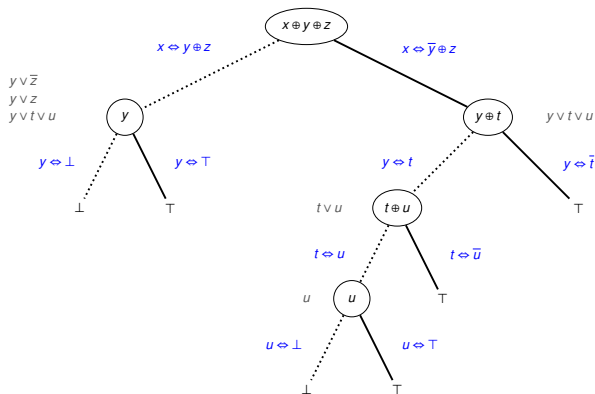
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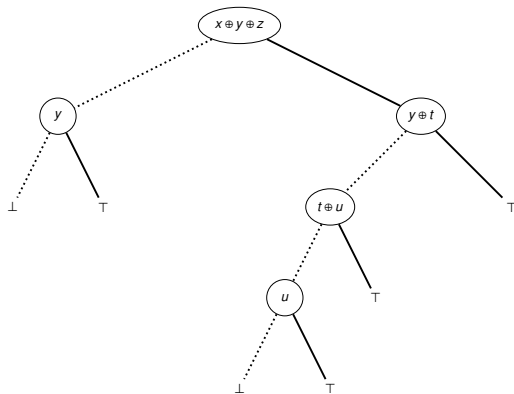
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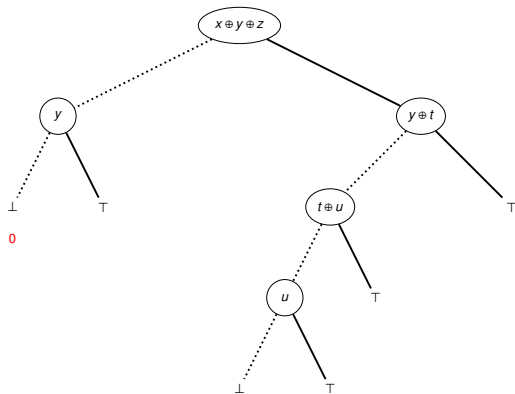
Generation and Model Counting

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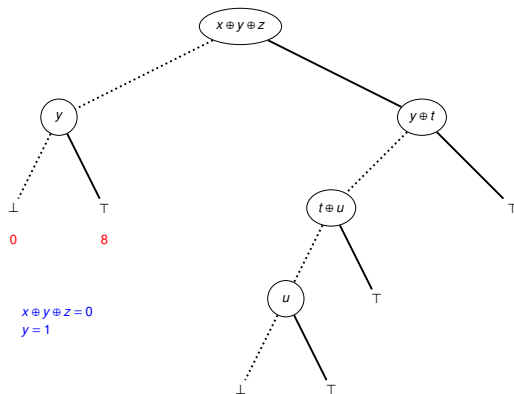
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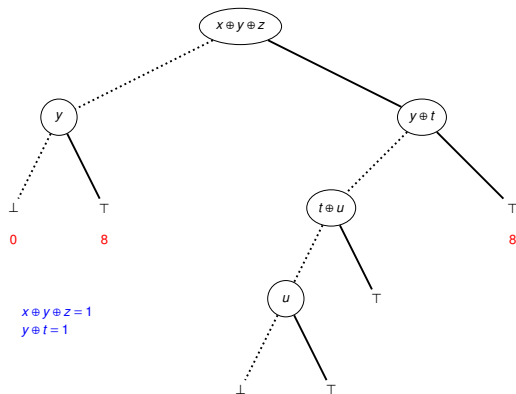
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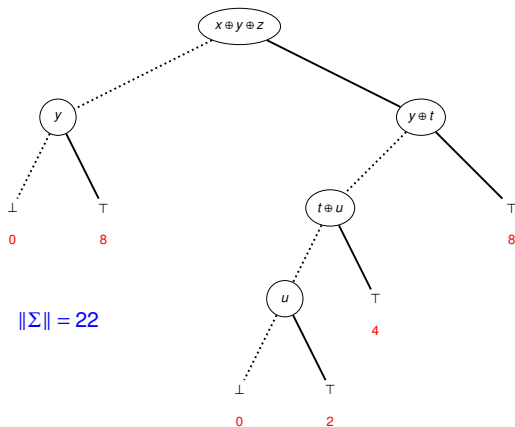
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Generation and Model Counting

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$$\|\Sigma\| = 22$$

KC Map for ADT: Queries

$\mathcal{L}$	CO	VA	CE	IM	EQ	SE	CT	ME
Circ	○	○	○	○	○	○	○	○
CNF	○	✓	○	✓	○	○	○	○
DNF	✓	○	✓	○	○	○	○	✓
ODNF	✓	✓	✓	✓	✓	✓	✓	✓
MODS	✓	✓	✓	✓	✓	✓	✓	✓
decision-DNNF	✓	✓	✓	✓	?	○	✓	✓
FBDD	✓	✓	✓	✓	○	○	✓	✓
DT	✓	✓	✓	✓	✓	✓	✓	✓
AFF	✓	✓	✓	✓	✓	✓	✓	✓
ADT	✓	✓	✓	✓	✓	✓	✓	✓

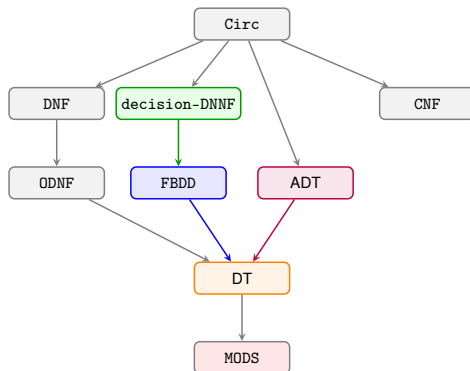
- **A Perfect Score:** Because ADT is fundamentally a tree structure that evaluates affine clauses, it completely inherits the polynomial-time query power of both DT and AFF.

KC Map for ADT: Transformations

$\mathcal{L}$	CD	FO	SFO	$\wedge C$	$\wedge BC$	$\vee C$	$\vee BC$	$\neg C$
Circ	✓	○	✓	✓	✓	✓	✓	✓
CNF	✓	○	✓	✓	✓	○	✓	○
DNF	✓	✓	✓	○	✓	✓	✓	○
ODNF	✓	○	○	○	✓	○	○	○
MODS	✓	✓	✓	○	✓	○	✓	○
decision-DNNF	✓	○	○	○	○	○	○	?
FBDD	✓	○	○	○	○	○	○	✓
DT	✓	○	✓	○	✓	○	✓	✓
AFF	✓	✓	✓	✓	✓	○	○	○
ADT	✓	○	✓	○	✓	○	✓	✓

- **The Baseline:** ADT mirrors the exact transformation profile of DT.
- **The Advantage:** Even though the profile is identical, ADT is **strictly more succinct** because the decision nodes can encode XOR relationships natively, avoiding exponential depth blow-ups.

Succinctness Hierarchy



- **decision-DNNF** and **ADT** are incomparable ($\nsubseteq_s$) to each other, representing two entirely different paradigms for pushing the limits of Knowledge Compilation!

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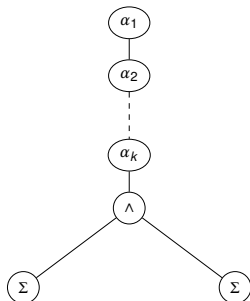
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Extended Affine Decision Tree (EADT)

- Just like we upgraded FBDD to decision-DNNF, we can take advantage of **Decomposability** to reduce the size of ADT.
- We introduce explicit $\wedge$ -nodes (and $\vee$ -nodes) which are **affine decomposable**.



Affine Decomposability

- $\forall i, \alpha_i$ is an affine clause.
- The $\wedge$ -node is standard decomposable:
 $Vars(\Phi) \cap Vars(\Psi) = \emptyset$.
- **Crucially:** For every α_i , it cannot share variables with *both* children simultaneously.

KC Map for EADT: Queries

$\mathcal{L}$	CO	VA	CE	IM	EQ	SE	CT	ME
Circ	○	○	○	○	○	○	○	○
CNF	○	✓	○	✓	○	○	○	○
DNF	✓	○	✓	○	○	○	○	✓
ODNF	✓	✓	✓	✓	✓	✓	✓	✓
MODS	✓	✓	✓	✓	✓	✓	✓	✓
decision-DNNF	✓	✓	✓	✓	?	○	✓	✓
FBDD	✓	✓	✓	✓	○	○	✓	✓
DT	✓	✓	✓	✓	✓	✓	✓	✓
AFF	✓	✓	✓	✓	✓	✓	✓	✓
ADT	✓	✓	✓	✓	✓	✓	✓	✓
EADT	✓	✓	✓	✓	?	○	✓	✓

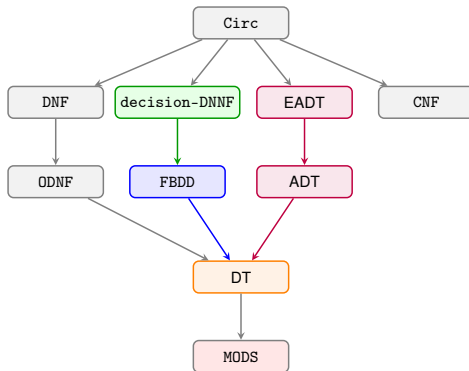
- **The Cost of Decomposability:** Just as moving from FBDD to decision-DNNF cost us Equivalence (**EQ**) and Sentential Entailment (**SE**), adding decomposable $\wedge$ -nodes to ADT forces EADT to lose those same properties!

KC Map for EADT: Transformations

$\mathcal{L}$	CD	FO	SFO	$\wedge C$	$\wedge BC$	$\vee C$	$\vee BC$	$\neg C$
Circ	✓	○	✓	✓	✓	✓	✓	✓
CNF	✓	○	✓	✓	✓	○	✓	○
DNF	✓	✓	✓	○	✓	✓	✓	○
ODNF	✓	○	○	○	✓	○	○	○
MODS	✓	✓	✓	○	✓	○	✓	○
decision-DNNF	✓	○	○	○	○	○	○	?
FBDD	✓	○	○	○	○	○	○	✓
DT	✓	○	✓	○	✓	○	✓	✓
AFF	✓	✓	✓	✓	✓	○	○	○
ADT	✓	○	✓	○	✓	○	✓	✓
EADT	✓	○	○	○	○	○	○	✓

- While standard decision-DNNF loses the guarantee of Negation (?), EADT **retains it** (✓)!

Succinctness Hierarchy



- decision-DNNF and EADT are incomparable ($\not\leq_s$) to each other, EADT is more succinct than ADT!